

For TSO, we extend the *CONWHILE* sequential commands:

$C \in Com ::= mfence$ [memory fence]

TSO Configurations:

- **Buffer**, as a FIFO sequence of delayed write labels: $b \in Buff \triangleq Seq < WLab > . WLab \triangleq \{(W, x, v) \mid x \in Loc \wedge v \in Val\}$.
- **Buffer Map**: $B \in BMap \triangleq Tid \rightarrow Buff . b = B(\tau)$: the buffer of τ .
- **TSO Configuration**: (P, S, M, B) .

TSO Transition Label: either an SC label, or a memory fence label *MF* for executing an *mfence*.